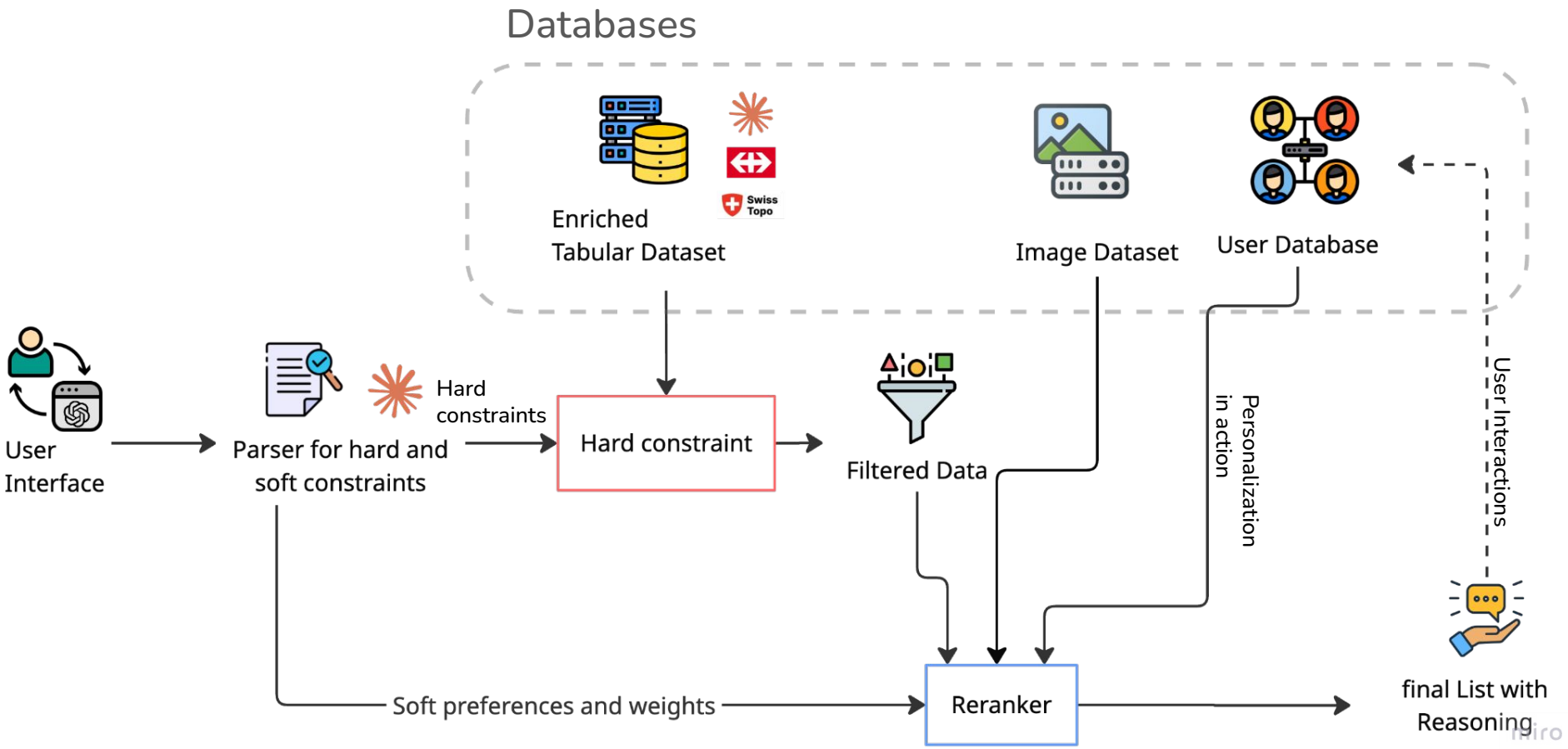
NestFinder: Search That Understands What You Actually Mean

Akmal Ashirmatov , Diego Zafferani, Jing Yan, Darya Zare


Databases
Enriched
Tabular Dataset
Swiss
Topo
Image Dataset
User Database
User
Interface
Parser for hard and
soft constraints
Hard
constraints
Hard constraint
Filtered Data
Personalization
in action
User Interactions
Soft preferences and weights
Reranker
final List with
Reasoning
miro

Data Enrichment

5 sources, 40+ features, zero labeling

Where?

- Municipality, district, canton, population density (swissBOUNDARIES3D)

How connected?

- Nearest stop, train station & Hauptbahnhof (SBB API)

What's nearby?

- 12 POI types (OpenStreetMap)

Fair price?

- CHF/m2 vs. municipality, district & canton avg

What's inside?

- 12 features from descriptions (Claude Haiku)

Query understanding

Raw query input

"3-room bright apartment in Zurich under 2800 CHF with balcony, close to public transport, ideally quiet"

Claude (LLM parser)

Structured JSON output prompt

Detects intent, intensity, hedge words, Swiss terms (3.5-Zimmer, ab sofort)

Hard filters

city: **Zurich**
rooms: **3.0 exactly**
max_price: **CHF 2800**
offer_type: **RENT**
Fail any -> listing excluded

Soft filters: intensity-weighted scoring

Key insight: `"must have parking"` scores **6x higher** than `"bright would be nice"` — intensity detected from language, not from explicit settings.

Claude identifies hedge words (ideally, would be nice, if possible) and certainty markers (must, essential, required), mapping them to weight multipliers 0.15 to 1.00.

Soft Signal

Soft Fact Extraction

OK 1296ms

Signals

bright 0.8

Extras

max_commute_minutes	30
commute_destination	–
preferred_min_area_sqm	–

Soft Filtering (Junk Removal)

OK 0ms

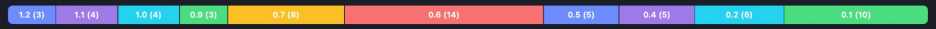
Before	62
After	62
Junk removed	0

Ranking

OK 43ms

Image bonus cap	0.550
Listings boosted by image	32
Max image bonus	0.550

Score distribution across all 62 candidates



#	FINAL	SOFT	Q+	IMG+	CITY	ROOMS	CHF	AREA	MATCHED SIGNALS	TITLE
1	1.21	0.53	0.17	0.51	Zürich	3	1,977	75 m²	bright +0.528	[Tauschwohnung] Teilsanierte 2 Zimmerwohnung im Herzen von Z details
2	1.17	0.53	0.12	0.52	Zürich	3	2,719	48 m²	bright +0.528	Wohnglück auf den ersten Blick details
3	1.16	0.53	0.15	0.48	Zürich	3	2,750	70 m²	bright +0.528	[Tauschwohnung] Wunderschöne Altbauwohnung am Idaplatz details
4	1.10	0.40	0.20	0.50	Zürich	3	1,910	70 m²	bright +0.4	Top renovierte 3-Zimmerwohnung mit allem Komfort, 4.OG, im H details
5	1.08	0.40	0.13	0.55	Zürich	3	2,560	74 m²	bright +0.4	Ruhige und helle 3.0-Zimmerwohnung an bester Lage im Kreis 7 details
6	1.06	0.40	0.17	0.49	Zürich	3	1,906	65 m²	bright +0.4	Helle 3-Zimmer-Wohnung mit Balkon, 1906.- in Witikon details
7	1.06	0.40	0.16	0.50	Zürich	3	1,906	65 m²	bright +0.4	3 Zimmer Wohnung details
8	1.03	0.40	0.15	0.48	Zürich	3	1,950	—	bright +0.4	3 Zimmer-Wohnung, 8051 Zürich details
9	1.02	0.40	0.12	0.50	Zürich	3	2,790	—	bright +0.4	ERSTVERMIETUNG nach Totalsanierung details
10	1.02	0.40	0.16	0.46	Zürich	3	2,020	—	bright +0.4	3.0 Zimmerwohnung in Zürich-Altstetten zu vermieten! details
11	1.01	0.40	0.16	0.45	Zürich	3	2,100	65 m²	bright +0.4	[Tauschwohnung] Kleiner gegen Grösser details
12	0.93	0.26	0.15	0.52	Zürich	3	2,273	68 m²	bright +0.264	stadtnah wohnen, im Zentrum von Oerlikon details
13	0.91	0.26	0.16	0.48	Zürich	3	1,841	65 m²	bright +0.264	[Tauschwohnung] 3Raum am Uetliberg mit Südbalkon, katzenfreu details
14	0.90	0.26	0.16	0.48	Zürich	3	2,073	69 m²	bright +0.264	moderne Wohnung in Witikon - Befristet bis 31.03.2027 details

Image RAG Service

Rerank candidate listings by visual similarity

"I am looking for …"

{
Listing a : 0.95
Listing b : 0.92
Listing c : 0.85
}

Ranking

1. **Extract Preferences**
   bright, quiet, balcony, public_transport, ...
2. **Assign Weights**
   each signal gets a weight value
3. **Calculate Score**
   weight × match_strength per signal
4. **Apply Quality Bonus**
   for stronger tie-breaking
5. **Boost by Images**
   match query images
6. **Boost by History**
   align with past preferences

Personlization in action

NestFinder | same query | cold vs warm | real demo output

"quiet apartment in Zurich,
must have parking,
balcony would be nice"

COLD

No history | flat weights | generic reasons

#1 Studio Apt — Weinbergstrasse CHF 1900
Zurich | 1.0r | Score: 2.200
quiet | balcony | weights x1.0
Matched soft filters. Flat weights, no personalization.

#2 Neubauwohnung in Zurich CHF 3080
Zurich | 3.5r | Score: 1.660
parking | balcony | weights x1.0
Matched soft filters. No preference signal.

#3 Modernes Wohnen im Guggachpark CHF 3910
Zurich | 4.5r | Score: 1.660
parking | balcony | weights x1.0
Matched soft filters. Flat weights.

#4 v Wohnung an ruhiger und zentraler Lage CHF 2330
Zurich | 3.0r | Score: 0.400
balcony only | no elevator signal
Will drop to #5 in warm — outranked by elevator affinity.

#5 ^ Wohnen im Trendquartier — 3.5 Zimmer CHF 3190
Zurich | 3.5r | Score: 0.400
balcony | elevator in description
Will rise to #4 in warm — elevator matches user affinity.

RANKING DELTA — COLD TO WARM

#1 Studio — Weinbergstrasse was #1 | score +0.145
#2 Neubauwohnung Zurich was #2 | score +0.120
#3 Modernes Wohnen Guggachpark was #3 | score +0.080
#4 ▲ Trendquartier was #5 | elevator boost
#5 ▼ Wohnung ruhiger Lage was #4 | no elevator

WARM

Profile applied | clicks, favorites, price range

#1 Studio Apt — Weinbergstrasse CHF 1900
Zurich | 1.0r | Score: 2.345 (+0.145) | clicked
quiet | balcony | preferred city | price CHF 1910-2073
Personalised: preferred city, price range, previously clicked.

#2 Neubauwohnung in Zurich CHF 3080
Zurich | 3.5r | Score: 1.780 (+0.120)
parking | balcony | preferred city boost
Personalised: preferred city boost applied.

#3 Modernes Wohnen im Guggachpark CHF 3910
Zurich | 4.5r | Score: 1.740 (+0.080)
parking | balcony | preferred city boost
Personalised: preferred city boost applied.

#4 ^ Wohnen im Trendquartier — 3.5 Zimmer CHF 3190
Zurich | 3.5r | Score: 0.520 (+0.120)
balcony | elevator match | preferred city
Rose from #5 — elevator preference in user profile matched.

#5 v Wohnung an ruhiger und zentraler Lage CHF 2330
Zurich | 3.0r | Score: 0.480
balcony | preferred city | no elevator
Dropped from #4 — no elevator signal, outranked by Trendquartier.

USER PROFILE APPLIED

Preferred city: **Zurich**
Price range: **CHF 1910-2073**
Feature affinity: **elevator**
Clicked: **listings 58963, 2671**
Favorited: **listing 2671**
Built from: **0 explicit inputs | behavior only**

Same query. Better ranking with personalization.

quiet apartment in Zurich, must have parking, balcony would be nice

COLD SEARCH

No user context

2.20

Top score

#1 Studio Apartment · Zürich

1.66

Runner-up

#2 Neubauwohnung

Top result

quiet • balcony

WARM SEARCH

Profile-aware ranking

2.35

Top score

#1 Studio Apartment · Zürich

1.78

Runner-up

#2 Neubauwohnung

Top result

quiet • balcony • parking fit

Zürich preference · Budget CHF 1910–2073

Likes elevator · Click/favorite history

Insight · Personalization resolves close matches using learned intent.

Thanks!